Practical - 14

Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition : Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

A. Theory

1. Are dashboard values loaded dynamically from the database?

Yes. All metrics on the admin dashboard (Active Students, Total Events, Total Registrations, Admin Accounts, and Audit Log count) are calculated dynamically using SQL queries (SELECT COUNT(*) FROM ...). The "Latest Registrations" table is also fetched dynamically using an INNER JOIN across registrations, students, and events.

2. Is access restricted to admin users?

Yes. Every administrative page checks $_SESSION['role'] === 'admin' at the very top of the script. If an unauthorized user (student or guest) tries to access the page directly via URL, they are immediately blocked with header("Location: access_denied.php"); exit();.

3. Are role-based menus working?

Yes. The modular navbar.php checks the session role:

When logged in as Admin: Displays Dashboard, Students, Events, Audit Logs, and Logout (Admin).

When logged in as Student: Displays Student Home, View Events, Try Admin Area, and Logout (Student).

When Guest: Displays Login only.

4. Are important admin actions recorded in audit_logs?

Yes. A dedicated audit_logs table records actions using record_audit_log($conn, $action, $details). Actions such as Admin Login, Logout, Announcement broadcasts, diagnostics, event addition/deletion, and role changes are stored with the user email, role, action type, description, client IP, and timestamp.

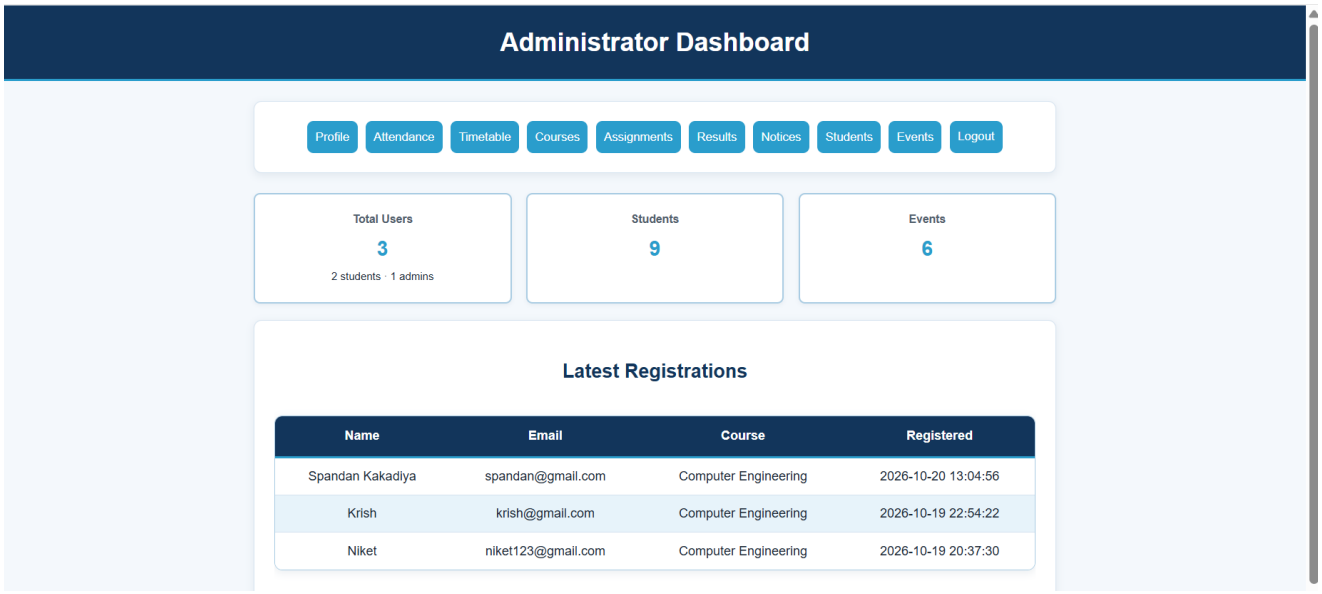
B. Implementation

Procedural Architecture: Built with beginner-accessible procedural PHP (session_start(), mysqli_query(), mysqli_fetch_assoc()).

RBAC: Enforced server-side using $_SESSION['role'] checks on all protected routes.

Dynamic Menu: Reusable navbar.php component utilizing basic if/elseif conditions to render menu options according to role.
Audit System: Centralized record_audit_log() helper inserting structured event trails into the database.

C. Evidence of Work (Screenshots)







ID	Name	Email	Course	Year / Division	Actions
STU1001	Aarav Shah	aarav.shah@example.com	Computer Engineering	Second Year / 1	Edit Delete
STU1002	Anaya Patel	anaya.patel@example.com	Information Technology	First Year / 2	Edit Delete
Het123	Het	het@gmail.com	Information Technology	Second Year / 1	Edit Delete
STU1004	Ira Kulkarni	ira.kulkarni@example.com	Computer Engineering	Second Year / 2	Edit Delete
STU1003	Kabir Desai	kabir.desai@example.com	Mechanical Engineering	Third Year / 1	Edit Delete
Krish123	Krish	krish@gmail.com	Computer Engineering	Second Year / 1	Edit Delete
Niket123	Niket	niket123@gmail.com	Computer Engineering	First Year / 1	Edit Delete
Pratik001	Pratik	pratik@gmail.com	Information Technology	2026-27 / 1	Edit Delete
Spandan037	Spandan Kakadiya	spandan@gmail.com	Computer Engineering	Second Year / 1	Edit Delete

D. GitHub Repository Link:

https://github.com/25dce037-tech/Student-Hub-Portal.git

E. Challenges Faced:

1. Session State Synchronization: Ensuring session destruction on logout completely resets both the PHP session array and browser cookie before redirecting.
2. Access Control Without Code Duplication: Creating a centralized access_denied.php handler and reusable navbar.php so role logic remains clean without repeating code across multiple pages.
3. Live Action Tracking: Ensuring audit logging occurs immediately alongside database operations (like INSERT/DELETE) without complicating the query flow.

F. Learning Outcomes:

1. Understood the fundamentals of Role-Based Access Control (RBAC) and server-side authorization beyond basic authentication.
2. Learned how to write SQL aggregate queries (COUNT(*)) and multi-table JOIN statements to feed real-time dashboards.
3. Learned how to implement Dynamic UI Navigation in PHP based on session variables.
4. Understood the importance of Audit Trails in web security for monitoring user activities and detecting unauthorized access attempts.

Faculty Remarks: